

A photograph of the Aurora Borealis (Northern Lights) over a snowy, rocky landscape. The aurora displays vibrant green and yellow-green curtains of light against a dark, star-filled night sky. The foreground shows dark, snow-covered ground with some rocks visible. The image is framed by a large, semi-circular teal overlay at the bottom.

# **PLANE CHANGE AND BI-ELLIPTIC TRANSFER WITH GMAT**

Simone Bisceglia





# ORBITAL MANEUVERS

As established in foundational orbital mechanics, defining the position and velocity vectors in an inertial frame uniquely determines the six associated orbital parameters.

Under Keplerian assumptions, five of these parameters are time-independent, while only the true anomaly varies with time.

Executing an orbital maneuver fundamentally alters the spacecraft's orbit, thereby modifying its orbital parameters.

Since specific parameters govern the orbit's shape and dimensions, while others define the orbital plane, maneuvers are classified based on the parameters targeted for modification:

- 1) Coplanar Maneuvers → modify only in-plane parameters, such as the **semi-major axis**  $\Delta a$ , **eccentricity**  $\Delta e$ , **argument of perigee**  $\Delta \omega$ , or **true anomaly**  $\Delta \nu$ .
- 2) Out of Plane Maneuvers → alter the orbital plane, defined by the inclination by **inclination**  $\Delta i$  and **right ascension of the ascending node**  $\Delta \Omega$ .
- 3) Mixed Maneuvers → couple both in-plane and out-of-plane parameter variations.



# ORBITAL MANEUVERS

The general orbital maneuvering problem can be formulated through Lambert's problem: given an initial position at a specific time, the objective is to reach a target position within a defined time interval.

This scenario is formally classified as a Space or Orbit Interception Problem.

A fundamental assumption in preliminary mission analysis is the impulsive maneuver model.

Under this framework, thrust application is considered instantaneous, resulting in an immediate velocity change in zero time, while the position vector remains strictly constant.

An instantaneous variation in the velocity vector fundamentally alters the trajectory, immediately yielding a new set of six orbital parameters and establishing a new orbital state.

Orbital transfers may require a single velocity variation or a sequence of multiple impulses, computed as:  $\Delta V_i$ ,  $i = 1, \dots, N$ .

For mission analysis and propellant budgeting, the total required effort is evaluated by the scalar sum of the velocity variations:  $\Delta V_{tot} = \sum_{i=1}^N |\Delta V_i|$ .

This computation relies exclusively on the algebraic sum of the vector norms rather than the vector sum, representing the total energetic cost of the maneuver sequence.

# Plane change maneuver

Delta-V calculation



# PLANE CHANGE MANEUVER



PLANE CHANGES ARE CRUCIAL MANEUVERS PERFORMED BY SPACECRAFT TO ALTER THEIR ORBITAL INCLINATION ALLOWING THEM TO REACH SPECIFIC TARGETS OR OPTIMIZE THEIR MISSIONS.



THIS PRESENTATION OUTLINES THE FUNDAMENTAL MECHANICS OF PLANE CHANGES, INCLUDING THE CORE PRINCIPLES OF ORBITAL INCLINATION AND ITS IMPACT ON SPACECRAFT DYNAMICS.



A SPECIFIC TEST CASE IS DEFINED TO DERIVE AN ANALYTICAL SOLUTION FOR THE REQUIRED  $\Delta V$ .

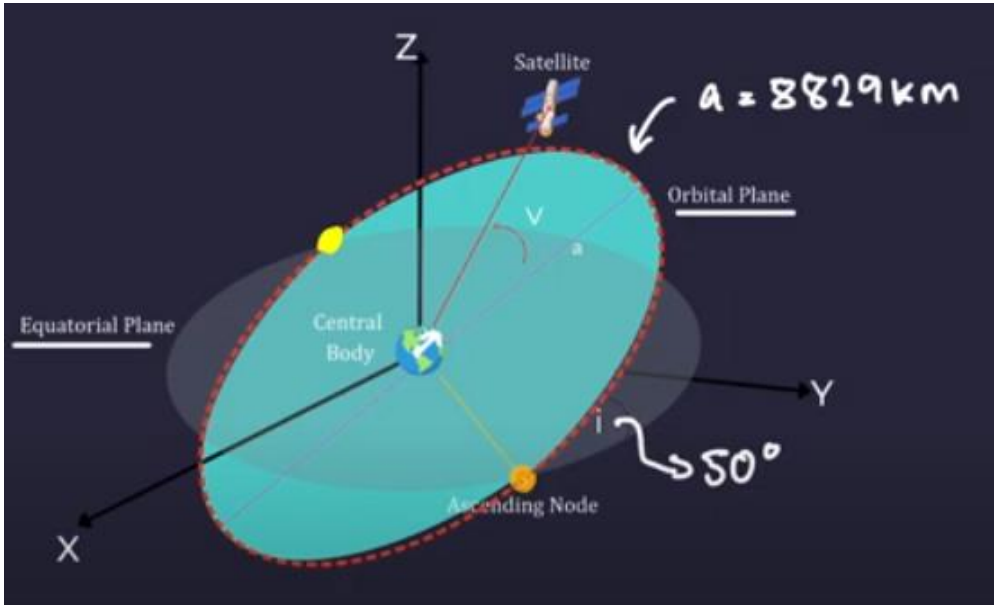
SUBSEQUENTLY, THE MANEUVER IS IMPLEMENTED AND SIMULATED USING THE GENERAL MISSION ANALYSIS TOOL (GMAT).



SPECIFICALLY, THE APPLICATION OF GMAT'S ITERATIVE SOLVERS IS EXAMINED TO PROVIDE A ROBUST NUMERICAL ESTIMATION AND OPTIMIZATION FOR THE MANEUVER EXECUTION.

# Delta-v calculation

## Analytical Solution



- ❑ Problem Statement: a spacecraft is in a circular Earth orbit at an altitude of 8829 km with an inclination of 50°.
- ❑ The objective is to perform a plane change maneuver to reach an equatorial orbit (0° inclination) while maintaining the current eccentricity and semi-major axis.
- ❑ To achieve a pure inclination change without altering the orbit's shape, a single-burn maneuver must be executed at the intersection between the initial and desired orbital planes, which typically corresponds to the ascending or descending node.

$$\Delta V_{pc} = 2V_{circ} * \sin \frac{\theta}{2}$$

$$V_{circ} = \sqrt{\frac{u}{r}} = \sqrt{\frac{G*m}{r}} = \sqrt{\frac{398600}{8829+6371}} = 5.121 \text{ km/s}$$

$$\Delta V_{pc} = 2 * 5.121 * \sin \frac{50}{2} = 4.323 \text{ km/s}$$



# DELTA-V CALCULATION

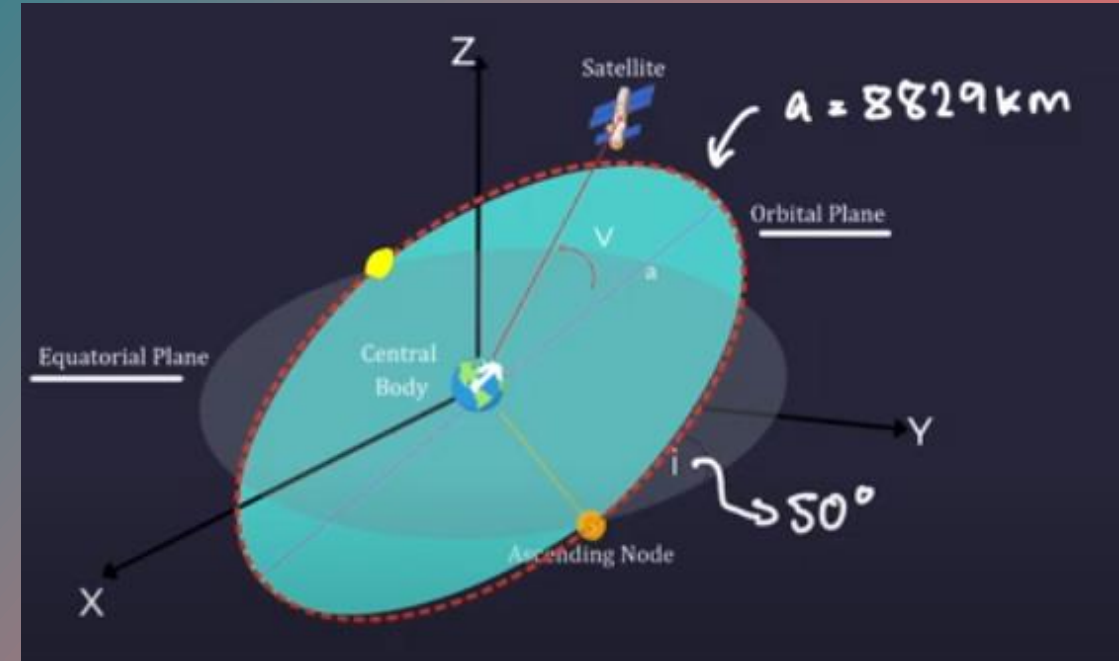
The initial step requires calculating the total  $\Delta V$  required for the plane change.

Since the initial and final orbits are circular and maintain the same altitude, the orbital velocity magnitude remains constant ( $V_i = V_f$ ).

To decrease the inclination by  $50^\circ$ , the velocity vector components must be adjusted.

Following the vector diagram of the maneuver, the required impulse is computed as follows:

$$\Delta \mathbf{V}_{pc} = [ (V_f * \cos\theta - V_i) \vec{i} + (V_f * \sin\theta) \vec{j} ] = [-1.829, -3.923] \text{ km/s.}$$

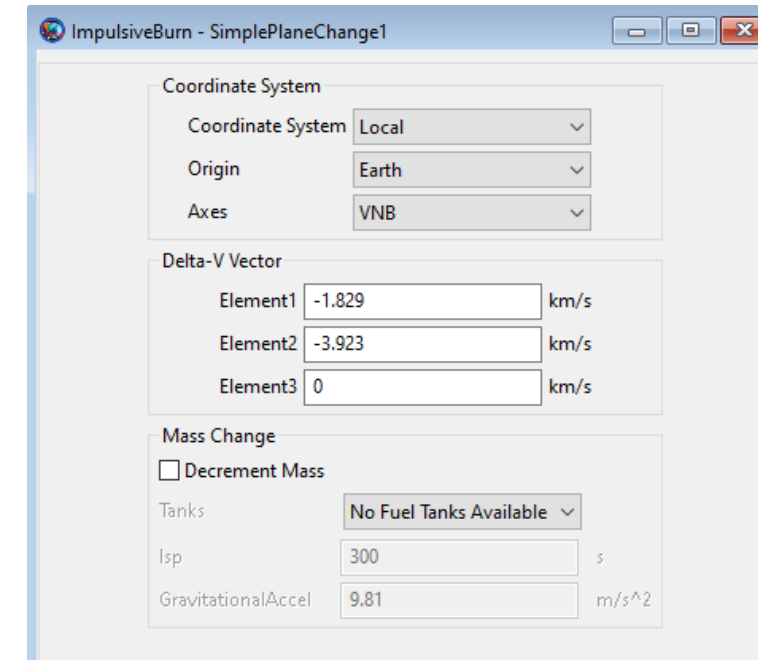
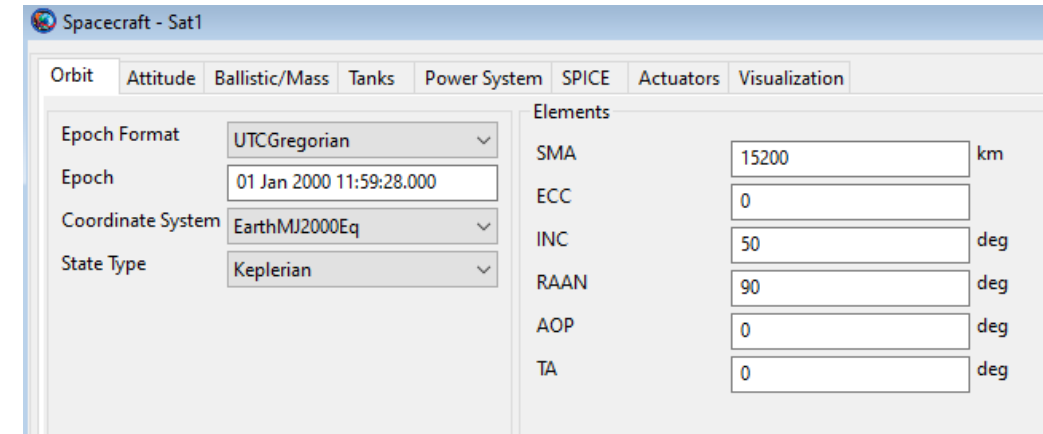
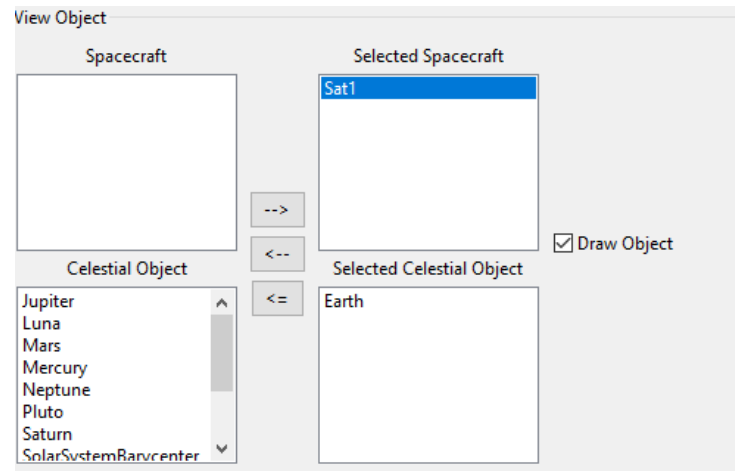


# GMAT SETUP

To validate the analytical solution and visualize the trajectory, the maneuver is implemented in the General Mission Analysis Tool (GMAT).

The setup process requires the following sequential steps:

1. **Spacecraft Initialization:** rename the default spacecraft object to sat1.
2. **Epoch Configuration:** set the epoch format to UTC Gregorian.
3. **State Definition:** change the state type to Keplerian and input the initial orbital parameters.
4. **Impulse Definition:** select the ImpulsiveBurn resource, as the maneuver is modeled as instantaneous, and input the analytically computed  $\Delta V$  components.
5. **Visualization Setup:** select the DefaultOrbitView, ensuring the active spacecraft is selected and the central celestial body is set to Earth.





# MISSION SEQUENCE

InitialOrbit

Propagators and Spacecraft

Propagate Mode: None ☐ Backwards Propagation ☐ Propagate STM ☐ Compute A-Matrix

| Propagator  | Spacecraft List |
|-------------|-----------------|
| DefaultProp | Sat1            |
|             |                 |
|             |                 |

Stopping Conditions

Stop Tolerance: 1e-07

| Parameter        | Condition |
|------------------|-----------|
| Sat1.ElapsedSecs | = 12000.0 |
|                  |           |
|                  |           |

GoToAN

Propagators and Spacecraft

Propagate Mode: None ☐ Backwards Propagation ☐ Propagate STM ☐ Compute A-Matrix

| Propagator  | Spacecraft List |
|-------------|-----------------|
| DefaultProp | Sat1            |
|             |                 |
|             |                 |

Stopping Conditions

Stop Tolerance: 1e-07

| Parameter     | Condition |
|---------------|-----------|
| Sat1.Earth.TA | = 180     |
|               |           |
|               |           |

Colors

☒ Override Color For This Segment Orbit Color  

SPC1

Burn SimplePlaneChange1

Spacecraft Sat1

☐ Backprop

FinalOrbit

Propagators and Spacecraft

Propagate Mode: None ☐ Backwards Propagation ☐ Propagate STM ☐ Compute A-Matrix

| Propagator  | Spacecraft List |
|-------------|-----------------|
| DefaultProp | Sat1            |
|             |                 |
|             |                 |

Stopping Conditions

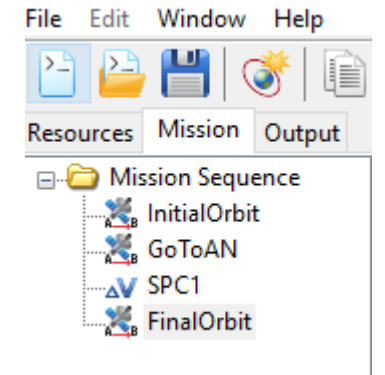
Stop Tolerance: 1e-07

| Parameter        | Condition |
|------------------|-----------|
| Sat1.ElapsedDays | = 1       |
|                  |           |
|                  |           |

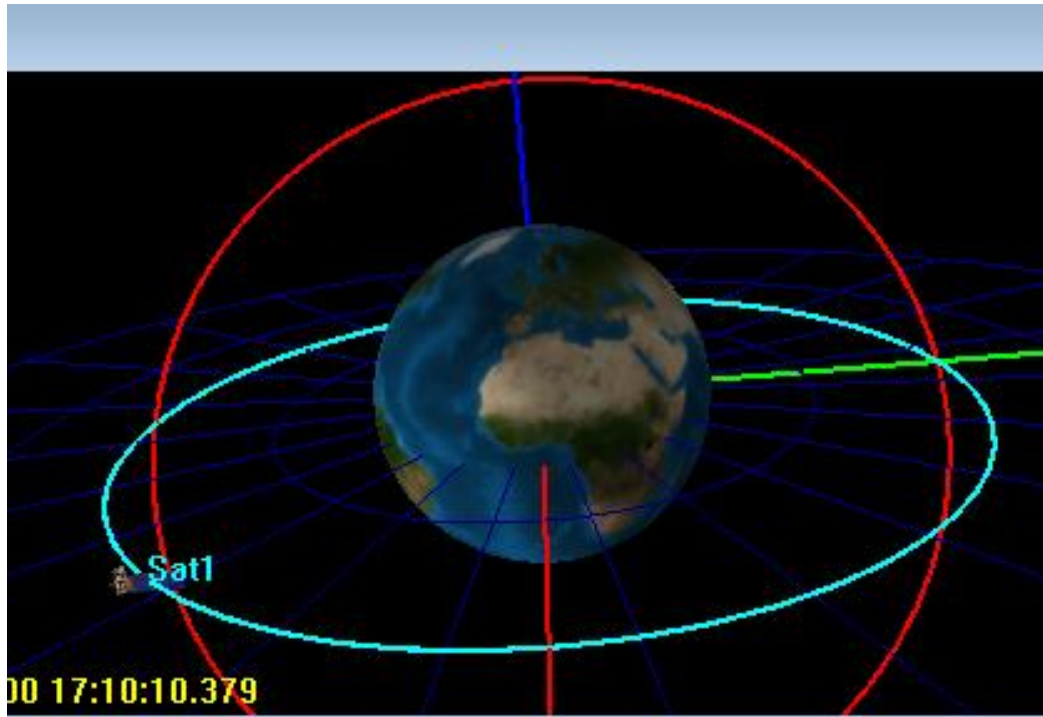
Colors

☒ Override Color For This Segment Orbit Color

# ORBIT AND GROUND TRACK VIEW → the sequence



ORBIT VIEW

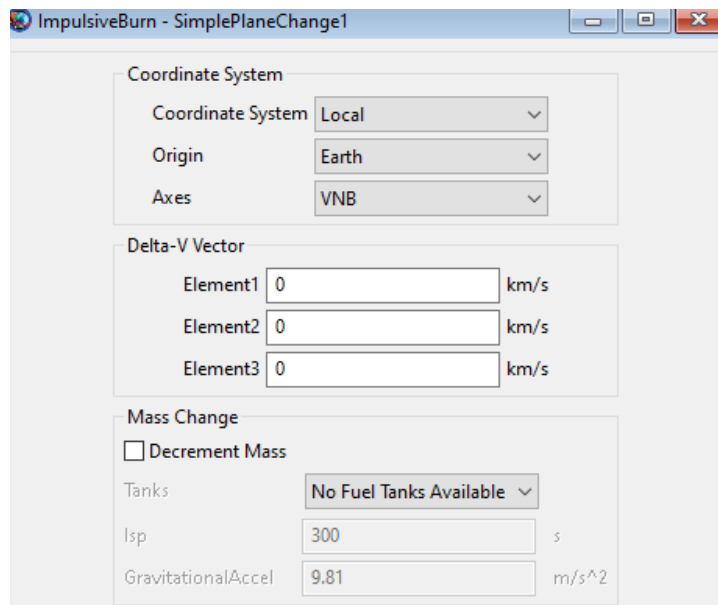


GROUND TRACK VIEW



# GMAT ITERATIVE METHOD $\rightarrow \Delta V$ ?

- Having validated the analytical solution, it is necessary to consider scenarios where the target inclination is defined, but the required  $\Delta V$  is unknown.
- Since analytical equations can be highly difficult to solve in certain situations, the GMAT solver must be employed.
- This requires the implementation of an iterative method by defining specific commands.
- First, a Boundary Value Solver must be introduced, which utilizes the Newton-Raphson numerical algorithm to converge upon a solution.
- Consequently, this necessitates the definition of both initial and boundary conditions.



ImpulsiveBurn - SimplePlaneChange1

Coordinate System

Coordinate System: Local

Origin: Earth

Axes: VNB

Delta-V Vector

Element1: 0 km/s

Element2: 0 km/s

Element3: 0 km/s

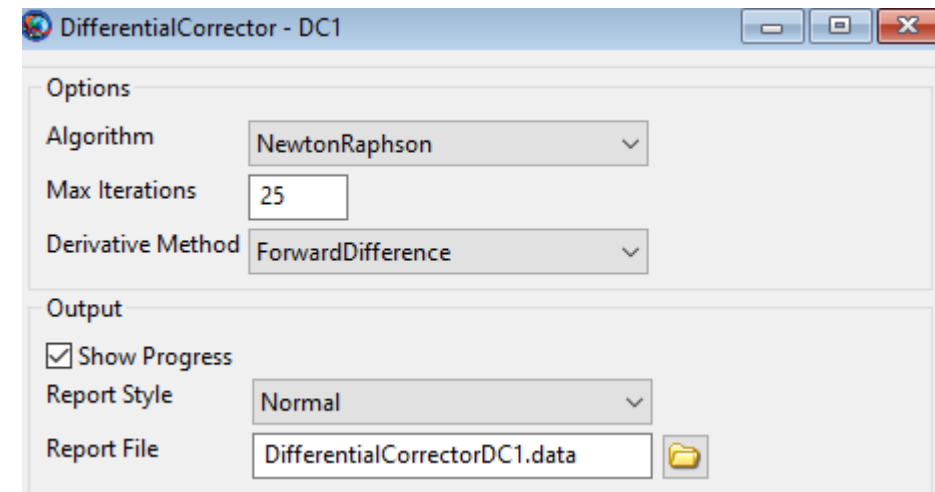
Mass Change

☐ Decrement Mass

Tanks: No Fuel Tanks Available

Isp: 300 s

GravitationalAccel: 9.81 m/s<sup>2</sup>



DifferentialCorrector - DC1

Options

Algorithm: NewtonRaphson

Max Iterations: 25

Derivative Method: ForwardDifference

Output

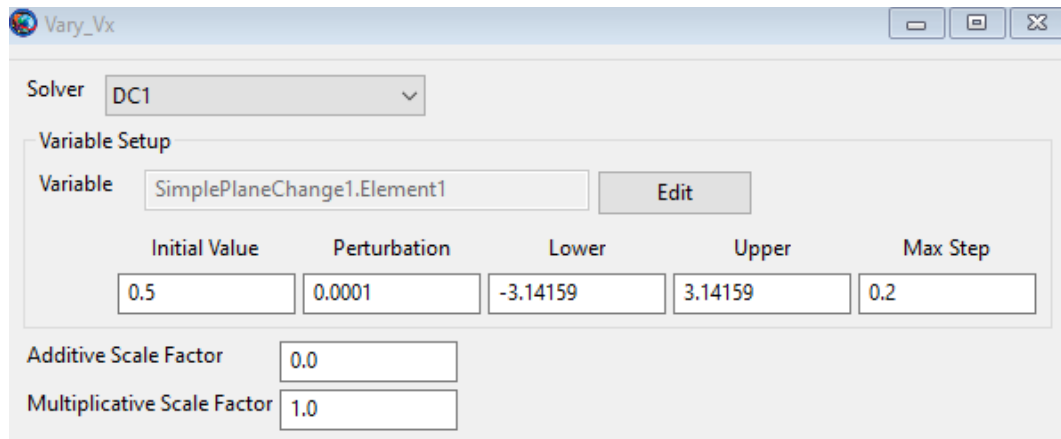
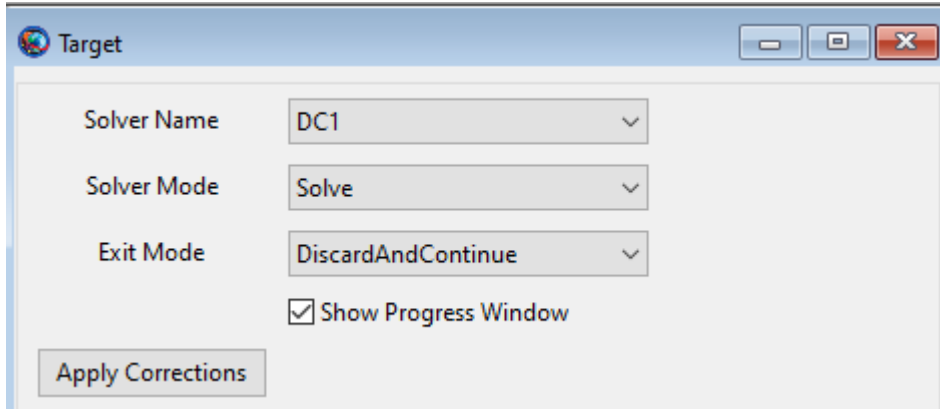
☒ Show Progress

Report Style: Normal

Report File: DifferentialCorrectorDC1.data



# GMAT ITERATIVE METHOD



- The mission sequence must be modified by defining a "Target", which establishes a point in the sequence to converge upon a solution for the desired parameter.
- A "Vary" command must be introduced to iteratively compute the velocity components along the three axes.
- Subsequently, the maneuver is executed by applying the impulse provided by the GMAT solver.
- Finally, an "Achieve" command must be introduced. The task of the "Achieve" command is to halt the function upon reaching the desired orbit, thereby ensuring the convergence of the solution.
- Consequently, a tolerance must be specified; this condition relates to the inclination and the magnitude of the final orbit, whose propagation must be represented as in the previous case

# GMAT ITERATIVE METHOD

Vary\_Vy

Solver: DC1

Variable Setup

Variable: SimplePlaneChange1.Element2 Edit

| Initial Value | Perturbation | Lower | Upper   | Max Step |
|---------------|--------------|-------|---------|----------|
| 0.5           | 0.0001       | -5    | 3.14159 | 0.2      |

Additive Scale Factor: 0.0

Multiplicative Scale Factor: 1.0

Vary\_Vz

Solver: DC1

Variable Setup

Variable: SimplePlaneChange1.Element3 Edit

| Initial Value | Perturbation | Lower | Upper   | Max Step |
|---------------|--------------|-------|---------|----------|
| 0.5           | 0.0001       | 0.0   | 3.14159 | 0.2      |

Additive Scale Factor: 0.0

Multiplicative Scale Factor: 1.0

SPC1

Burn: SimplePlaneChange1

Spacecraft: Sat1

☐ Backprop

AchieveINC

Solver: DC1

Goal: Sat1.EarthMJ2000Eq.INC Edit

Value: 0 Edit

Tolerance: 2 Edit

AchieveECC

Solver: DC1

Goal: Sat1.Earth.ECC Edit

Value: 0 Edit

Tolerance: 0.1 Edit

# X-Y PLOTS → convergency of the solution

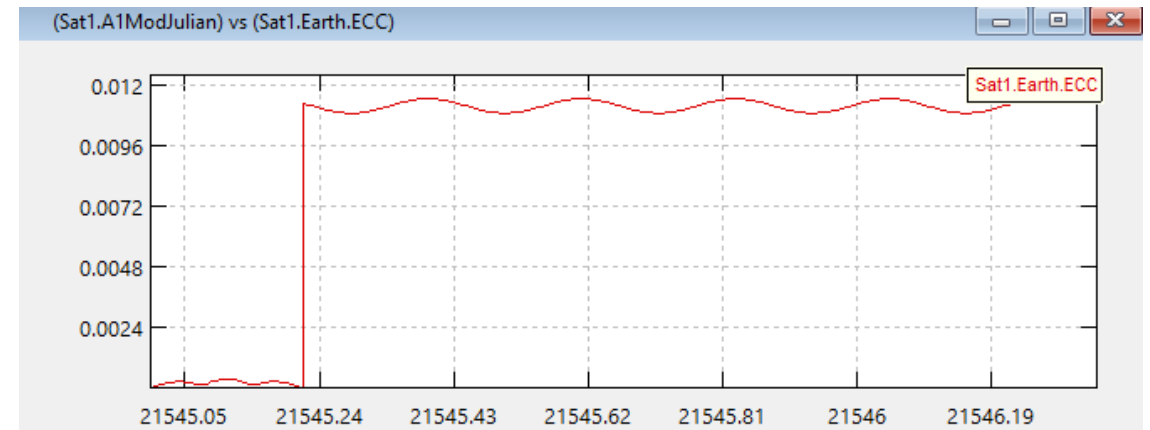
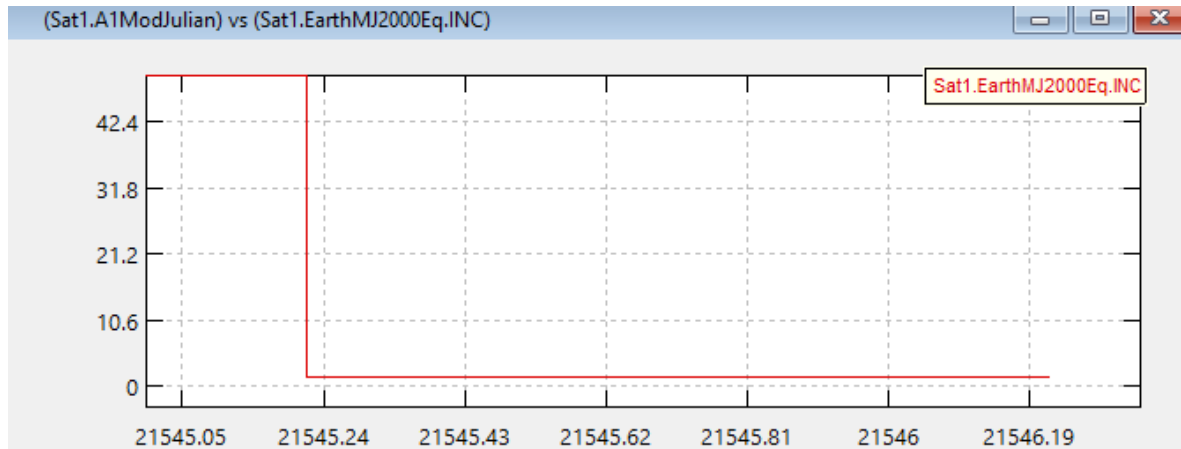
Solver Window - Target 'SPC1' DC1 {SolveMode = Solve, ExitMode = DiscardAndContinue, ShowProgressW...

| Control Variable            | Current Value       | Last Value          | Difference             |
|-----------------------------|---------------------|---------------------|------------------------|
| SimplePlaneChange1.Element1 | -1.711492838084512  | -1.711492838084512  | 2.220446049250313e-16  |
| SimplePlaneChange1.Element2 | -3.835522225293845  | -3.835522225293845  | -4.440892098500626e-16 |
| SimplePlaneChange1.Element3 | 0.05338508239436121 | 0.05338508239436121 | 0                      |

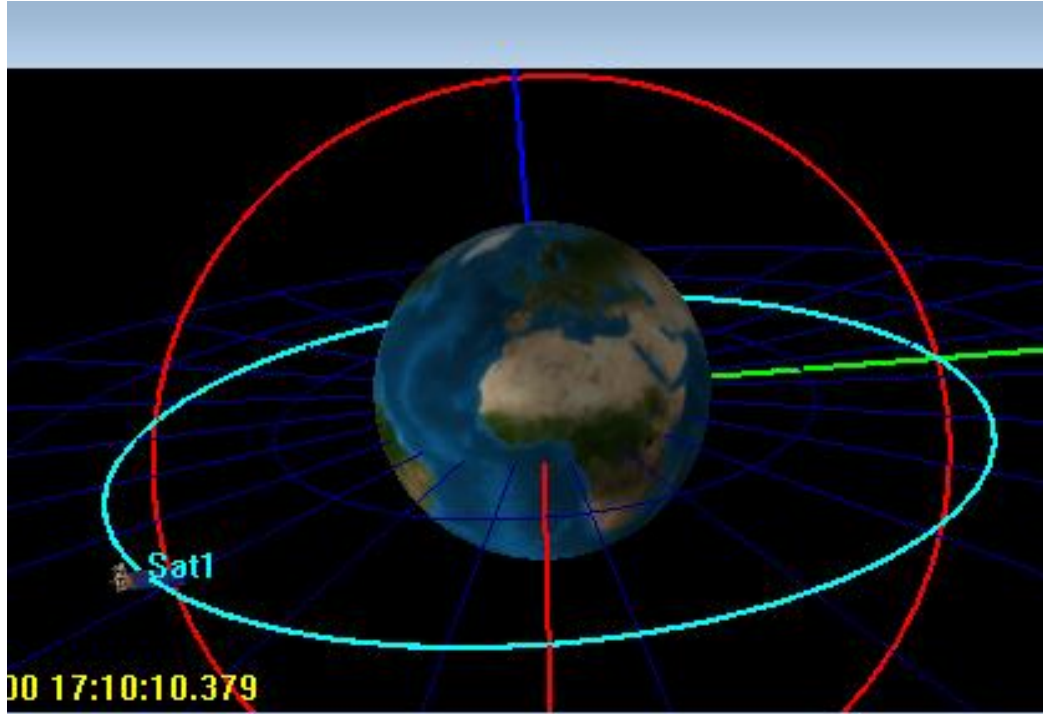
| Constraints                | Desired | Achieved            | Difference          |
|----------------------------|---------|---------------------|---------------------|
| (=) Sat1.EarthMJ2000Eq.INC | 0       | 1.633906799388814   | 1.633906799388814   |
| (=) Sat1.Earth.FCC         | 0       | 0.01128011433339654 | 0.01128011433339654 |

**CONVERGED**

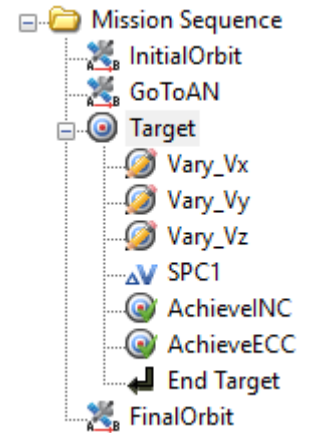


# ORBIT AND GROUND TRACK VIEW → the sequence

ORBIT VIEW



GROUND TRACK VIEW





# Bi-elliptic transfer on a GEO orbit

Description of the maneuver and the visualization on GMAT

# BI-ELLIPTIC TRANSFER

## CIRCULAR ORBITS: CHANGE OF RADIUS 3 $\Delta V$ s STRATEGY $\rightarrow$

The transfer orbit, in this case, is called Bi-Elliptic Transfer.

We have circular orbits, and we want to increase the radius, passing from the lower one ( $r_1$ ) to the higher one ( $r_2$ ).

As we want to use three tangential impulses, we need more than just one ellipse:

- 1) In the initial point we apply the first tangential  $\Delta V$ , so increasing the velocity we increase the semi-major axis but arriving beyond the final orbit:  $\Delta V_1 > 0 \Rightarrow c_1 \rightarrow tr_1$ ;
- 2) After half of the period on the first transfer ellipse ( $tr_1$ ) we apply our second tangential impulse at apogee to tune an ellipse with the opposite point on the final orbit:  $\Delta V_2 > 0 \Rightarrow tr_1 \rightarrow tr_2$ ;
- 3) As we arrive at the point on the final orbit, we apply the final impulse to circularize the orbit, but in this case, we perform an impulse to reduce the semi-major axis, so  $\Delta V_3$  is negative:  $\Delta V_3 < 0 \Rightarrow tr_2 \rightarrow c_2$ .

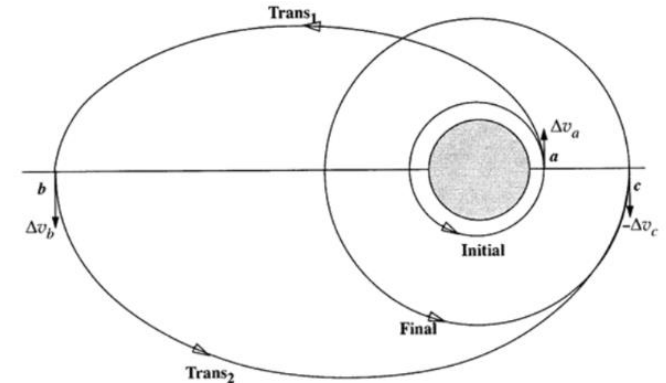
$$a_{tr1} = \frac{r_1 + r_b}{2}$$

$$a_{tr2} = \frac{r_2 + r_b}{2}$$

$$\Delta V_1 = V_{p,tr1} - V_{c1} = \sqrt{\mu \left( \frac{2}{r_1} - \frac{1}{a_{tr,1}} \right)} - \sqrt{\frac{\mu}{r_1}},$$

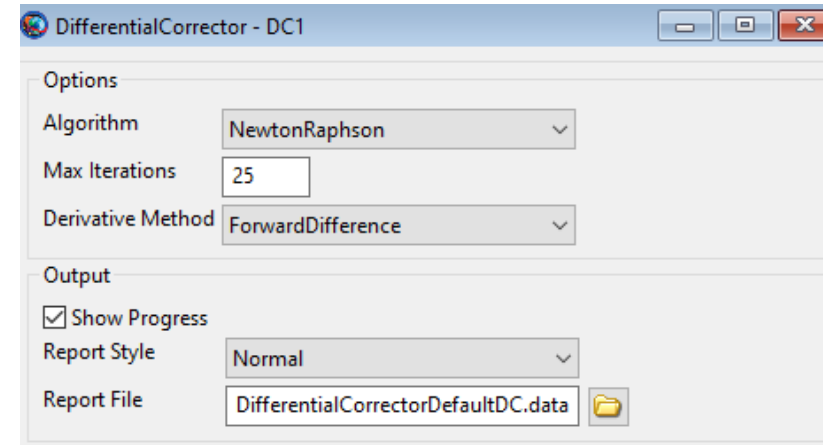
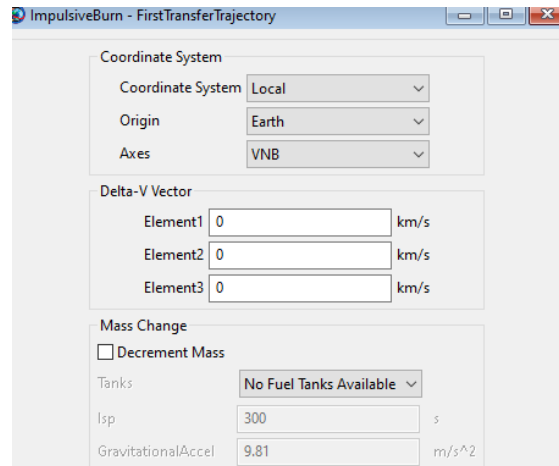
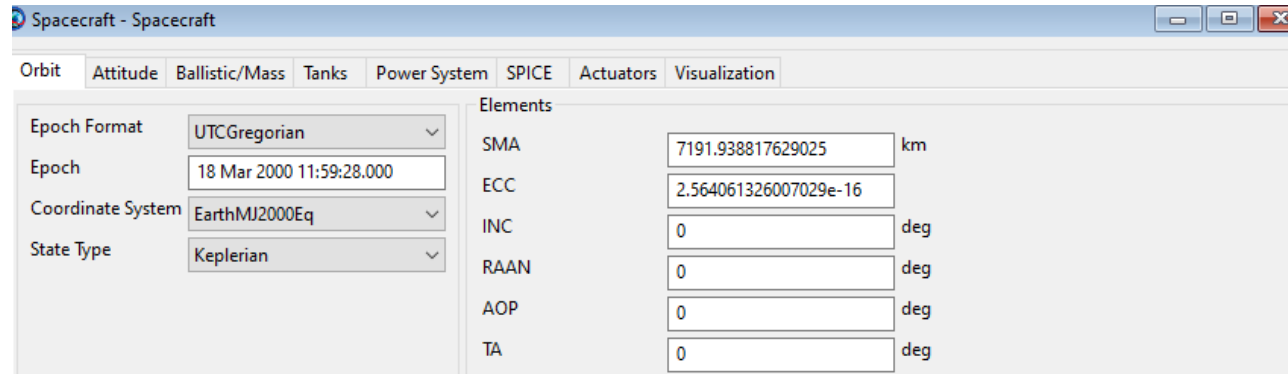
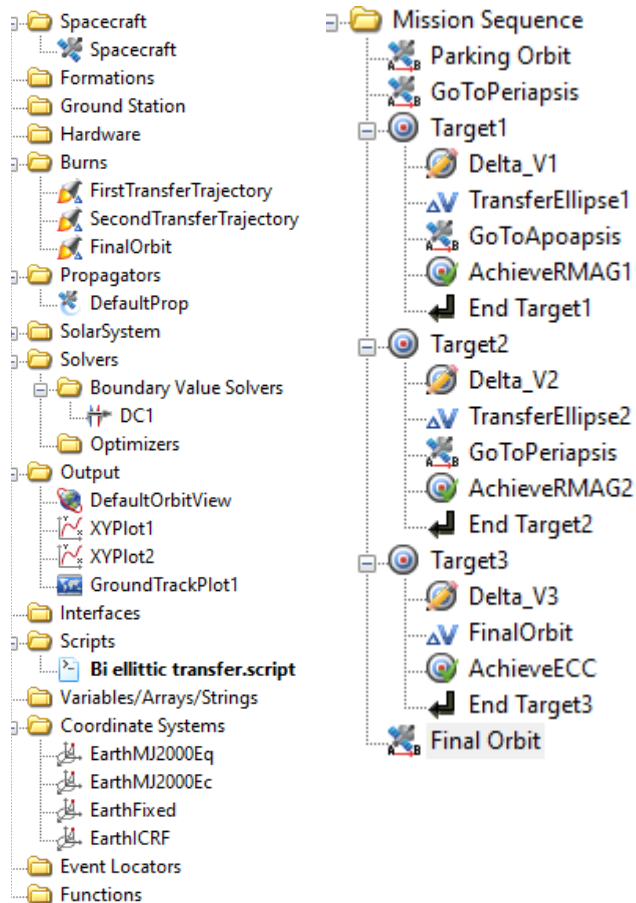
$$\Delta V_2 = V_{a,tr2} - V_{a,tr1} = \sqrt{\mu \left( \frac{2}{r_b} - \frac{1}{a_{tr,2}} \right)} - \sqrt{\mu \left( \frac{2}{r_b} - \frac{1}{a_{tr,1}} \right)},$$

$$\Delta V_3 = V_{c2} - V_{p,tr2} = \sqrt{\frac{\mu}{r_2}} - \sqrt{\mu \left( \frac{2}{r_2} - \frac{1}{a_{tr,2}} \right)}.$$



# GMAT SETUP

- Consistent with the GMAT operational workflow and the previous maneuver, the initial step requires defining the orbital parameters.
- It must be noted that the orbital plane remains unchanged; only the semi-major axis of the circular orbit (zero eccentricity) is modified.
- Consequently, the required  $\Delta V$  impulse differs, being purely tangential rather than rotational, thereby necessitating a modification of the mission sequence.



# X-Y PLOTS → convergency of the solution

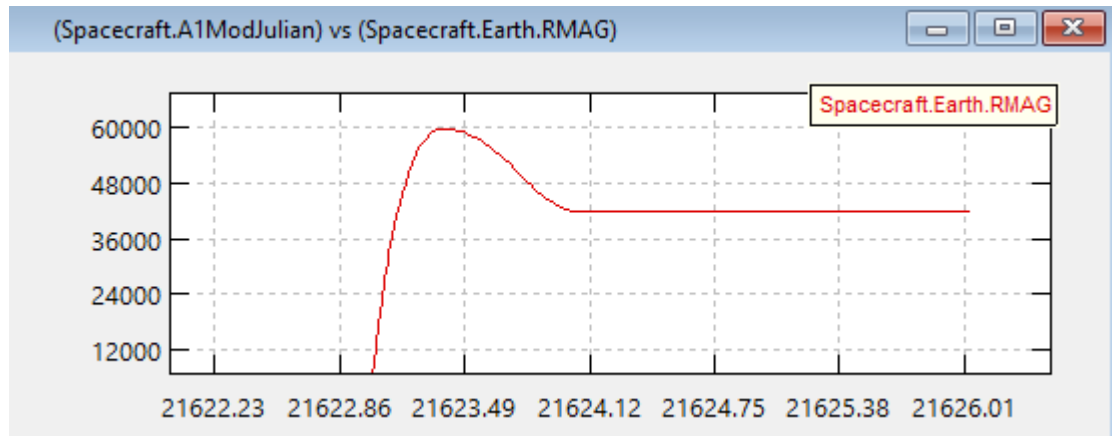
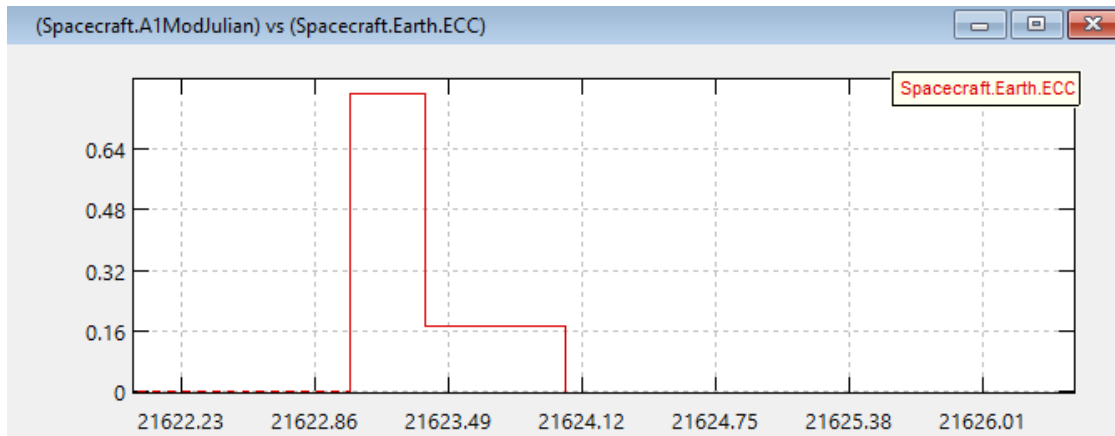
| Solver Window - Target DC1 {SolveMode = Solve, ExitMode = DiscardAndContinue, ShowProgressWindow ...} |                   |                   |                        |
|-------------------------------------------------------------------------------------------------------|-------------------|-------------------|------------------------|
| Control Variable                                                                                      | Current Value     | Last Value        | Difference             |
| FirstTransferTrajectory.Element1                                                                      | 2.501721987759553 | 2.501721987759553 | -4.440892098500626e-16 |
| Constraints                                                                                           | Desired           | Achieved          | Difference             |
| (==) Spacecraft.Earth.RMAG                                                                            | 60000             | 60000.07126737489 | 0.07126737488579238    |
| CONVERGED                                                                                             |                   |                   |                        |

| Solver Window - Target DC1 {SolveMode = Solve, ExitMode = DiscardAndContinue, ShowProgressWindow ...} |                   |                   |                      |
|-------------------------------------------------------------------------------------------------------|-------------------|-------------------|----------------------|
| Control Variable                                                                                      | Current Value     | Last Value        | Difference           |
| SecondTransferTrajectory.Element1                                                                     | 1.150281239622864 | 1.150281239622864 | 0                    |
| Constraints                                                                                           | Desired           | Achieved          | Difference           |
| (==) Spacecraft.Earth.RMAG                                                                            | 42165             | 42165.00550120173 | 0.005501201725564897 |
| CONVERGED                                                                                             |                   |                   |                      |

| Solver Window - Target DC1 {SolveMode = Solve, ExitMode = DiscardAndContinue, ShowProgressWindow ...} |                    |                       |                       |
|-------------------------------------------------------------------------------------------------------|--------------------|-----------------------|-----------------------|
| Control Variable                                                                                      | Current Value      | Last Value            | Difference            |
| FinalOrbit.Element1                                                                                   | 0.2570971258183061 | 0.2570971258183061    | 0                     |
| Constraints                                                                                           | Desired            | Achieved              | Difference            |
| (==) Spacecraft.Earth.ECC                                                                             | 0                  | 0.0003442560560854543 | 0.0003442560560854543 |
| CONVERGED                                                                                             |                    |                       |                       |

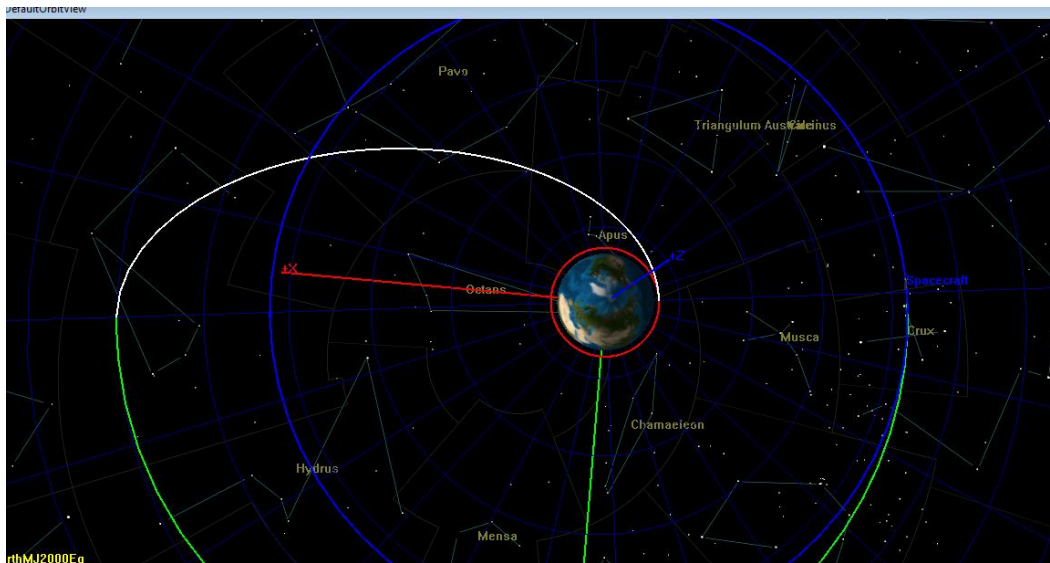


# X-Y PLOTS → convergency of the solution

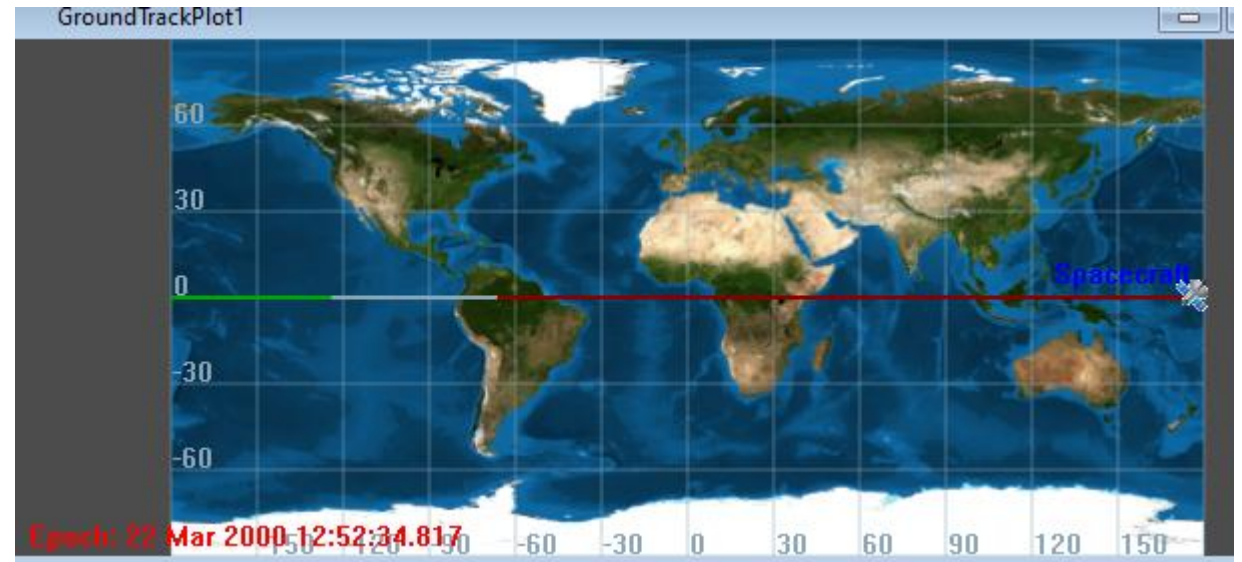


# ORBIT AND GROUND TRACK VIEW

ORBIT VIEW



GROUND TRACK VIEW



# Summary

This presentation has analyzed two distinct types of orbital maneuvers:

- ✓ **Plane Change Maneuver** (Out-of-Plane) → the analytical problem was initially solved, followed by the implementation of the solution in GMAT, which included an examination of the iterative numerical approach.
- ✓ **Coplanar Maneuver** → specifically, the Bi-Elliptic transfer to modify the orbital radius, evaluated by employing the iterative method provided by the GMAT solver.





**Thank You for the  
attention**